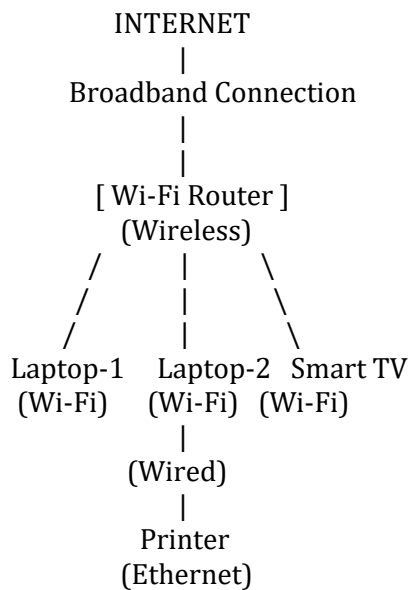


N+ - Network security, Maintenance and Troubleshooting procedures

❖ Build a SOHO network

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

ANS :



• Connection Types

Device	Connection Type
Broadband to Router	Wired
Laptop 1 to Router	Wireless (Wi-Fi)
Laptop 2 to Router	Wireless (Wi-Fi)
Smart TV to Router	Wireless (Wi-Fi)
Printer to Router	Wired (Ethernet Cable)

2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint : Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

ANS :

Component	Suggested Brand/Model
Broadband Internet Connection	JioFiber / Airtel Xstream Fiber
Wi-Fi Router	TP-Link Archer C6 AC1200
Laptop 1	HP 15s Laptop
Laptop 2	Lenovo IdeaPad Slim 3
Smart TV	Samsung 43" Smart TV
Printer	HP LaserJet MFP 136a
Ethernet Cable (Cat6)	D-Link Cat6 LAN Cable
Power Adapter	Included with Devices
Wi-Fi Extender (Optional)	TP-Link RE205 AC750
Network Switch (Optional if more wired devices needed)	TP-Link TL-SG105 5-Port Gigabit Switch

- **Why Optional Devices?**

- **Wi-Fi Extender:** Used if Wi-Fi signal is weak in some rooms.
- **Network Switch:** Used when additional wired devices need network connectivity.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

ANS :

- **Step 1: Connect Router**

1. Connect broadband cable to WAN port of router.
2. Connect laptop to router using Wi-Fi or LAN cable.
3. Power ON the router.

- **Step 2: Access Router Settings**

1. Open web browser.
2. Enter router IP:
3. 192.168.1.1 or 192.168.0.1
4. Login using default username and password.

- **Step 3: Create Unique SSID**

1. Open Wireless Settings.
2. Enter SSID:
3. Harsh_SOHO_Network
4. Save settings.

- **Step 4: Enable WPA2 Security**

1. Go to Wireless Security.
2. Select:
3. WPA2-PSK
4. Create strong password:
5. Harsh@2026WiFi
6. Save settings.

- **Step 5: Change Default Admin Password**

1. Open Administration Settings.
2. Change default password.
3. Example:
4. Username: admin
5. Password: HarshAdmin@2026
6. Save and reboot router.

- **Step 6: Verify Connection**

1. Connect all devices to Wi-Fi.
2. Open websites to confirm internet access.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

ANS :

- **Network Details**

- Router IP: 192.168.1.1
- Subnet Mask: 255.255.255.0
- DHCP Range: 192.168.1.100 – 192.168.1.200

- **Device Address Table**

Device	MAC Address	IP Address	Type
Wi-Fi Router	AA:BB:CC:DD:EE:01	192.168.1.1	Gateway
Laptop 1	00:1A:2B:3C:4D:11	192.168.1.101	DHCP
Laptop 2	00:1A:2B:3C:4D:12	192.168.1.102	DHCP
Smart TV	00:1A:2B:3C:4D:13	192.168.1.20	Static
Printer	00:1A:2B:3C:4D:14	192.168.1.10	Static

- **Static IP Configuration**

Printer

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

- Gateway: 192.168.1.1

Smart TV

- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

ANS :

- **Problem :** Laptop 1 cannot access the internet.

- **Cause 1: Wi-Fi Not Connected**

Diagnosis

1. Check Wi-Fi icon.
2. Verify correct SSID selected.
3. Confirm password entered correctly.

Solution

- Reconnect to Wi-Fi.
- Forget network and reconnect.
- Restart laptop Wi-Fi adapter.

- **Cause 2: Incorrect IP Address**

Diagnosis

- Open Command Prompt:
- ipconfig
- Check whether IP address is assigned.

Solution

- Renew IP address:
- ipconfig /release
- ipconfig /renew
- Ensure DHCP is enabled.

- **Cause 3: Router or Internet Service Issue**

Diagnosis

- Check router Internet LED.
- Test internet on another device.
- Ping router:
- ping 192.168.1.1

Solution

- Restart router.
- Check broadband connection.
- Contact ISP if outage exists.

- **Cause 4: DNS Configuration Problem**

Diagnosis

- ping 8.8.8.8
- If successful but websites do not open, DNS may be the issue.

Solution

- Set DNS manually:
- Primary DNS: 8.8.8.8
- Secondary DNS: 8.8.4.4

- **Cause 5: Firewall or Antivirus Blocking Access**

Diagnosis

- Temporarily disable firewall/antivirus and test connection.

Solution

- Update antivirus.
- Allow browser and network services through firewall.